Does Change-Awareness Help at Small Scale? A Critical Replication of CACo

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***Abstract*—CACo (Mall et al., CVPR 2023) learns satellite-image features with a contrastive loss that treats images of the same place taken years apart as negatives, but only when the place appears to have changed. We test its claims in two regimes. Using the authors' code, we pretrain MoCo v2, SeCo, a long-term-contrast variant (TeCo) and CACo on 2,000 Sentinel-2 locations. We then probe our SeCo, TeCo and CACo encoders, the authors' four released checkpoints (trained on ten times as many locations), ImageNet features and a random network under one fixed protocol, on EuroSAT and on UC Merced, an aerial dataset the paper does not use. With the released checkpoints, the paper's component ablation reproduces: on EuroSAT, long-term contrast adds 1.5 points and change-awareness 1.1 points. At our scale, change-awareness adds no detectable accuracy ($+0.6$, 95% CI $[-0.3, 1.4]$), and the long-term variant's large gain coincides with twice as many optimizer steps. Under a frozen-backbone probe, ImageNet features match the best satellite encoder on EuroSAT and lead every satellite encoder on UC Merced. Changing the probe protocol alone moves the same checkpoints by up to 8.7 points.**

I. Introduction

Earth-observation satellites revisit every location on the planet every few days, but almost none of this imagery is labelled. Self-supervised pretraining can turn the unlabelled archive into features that transfer to land-cover mapping, change detection and similar tasks with few labels. Satellite imagery offers a signal that natural images lack: the same place seen at many times. Images a few months apart differ mainly by season, while images years apart may show real, permanent change, such as a new road or neighbourhood (Fig. 1).

CACo [1] builds on this observation. It keeps SeCo's seasonal invariance [2] but contrasts images taken four years apart, and it estimates during training which locations changed so that unchanged places are not pushed apart. The paper reports consistent gains over SeCo and an ablation in which each of its components adds accuracy.

We ask whether these claims hold up under independent evaluation, and whether change-awareness still matters when pretraining data is scarce. Our contributions are:

- We pretrain four objectives (MoCo v2, SeCo, TeCo, CACo) with the authors' implementation on a 2,000-location archive, and release the SeCo, TeCo and CACo encoders, all training logs and our evaluation code.
- We evaluate nine encoders, including the authors' released checkpoints, under one protocol with five probe seeds and paired significance tests, on EuroSAT and on UC Merced, a 0.3 m aerial dataset that serves as a distribution-shift stress test.
- We verify five quantitative claims from the paper. Its component ablation reproduces with the released checkpoints, but change-awareness gives no measurable benefit at small scale.
- We document details that change how the paper's numbers should be read: SeCo receives half the optimizer steps of TeCo and CACo, the change mask is partial, CACo and TeCo also sample dates differently, and the probe protocol moves the same pretrained checkpoints by up to 8.7 points.

II. Literature Review

Contrastive self-supervised learning. Contrastive methods learn an encoder by pulling two views of the same image together and pushing views of different images apart under the InfoNCE objective [3]. SimCLR [4] draws negatives from large batches; MoCo and MoCo v2 [5], [6] keep a queue of negative keys produced by a slowly updated momentum encoder, which decouples the number of negatives from the batch size. All methods we study are built on MoCo v2 with a ResNet-18 [7] backbone.

Self-supervision for remote sensing. Satellite archives offer free structure that natural-image collections lack. Tile2Vec [8] uses spatial proximity and GSSL [9] uses geography and time. Seasonal Contrast (SeCo) [2] treats images of one location taken a few months apart as positives and learns three embedding subspaces: one invariant to both season and augmentation, one invariant to season only, and one invariant to augmentation only. SatMAE [10] instead pretrains vision transformers with masked autoencoding on temporal and multispectral data.

CACo. Mall et al. [1] extend SeCo with a second, distant time period. Images of the same location four years apart are used as negatives, so features become sensitive to permanent change. Because most places do not change, CACo estimates per location whether a change happened and keeps the distant image as a negative only when it did. The paper reports gains over SeCo on EuroSAT [11], BigEarthNet [12], change detection and segmentation, and an ablation (its Table 6) in which better geographic sampling, long-term contrast and change-awareness each add accuracy.

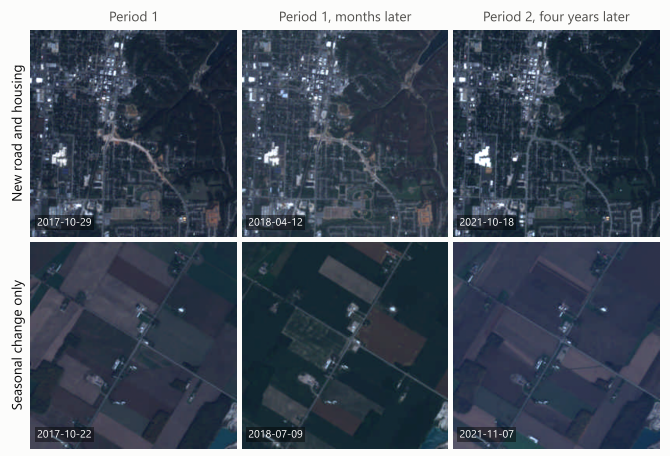


Fig. 1. Two locations from our pretraining data. Top: a road and a new subdivision are built between the periods. Bottom: fields change colour with the seasons but not structurally. CACo aims to be invariant to the bottom kind of change and sensitive to the top kind. Contains modified Copernicus Sentinel data (2017–2021).

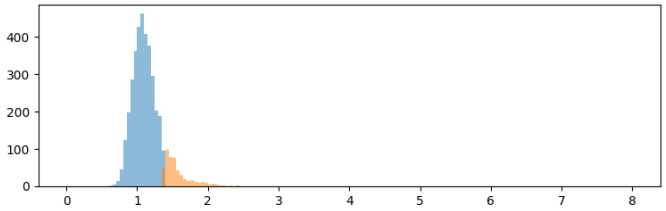


Fig. 2. CACo's change estimates during our pretraining run: histogram of the smoothed ratio of long-term to short-term feature distance over training items (x: ratio, y: count). Colours are the two Gaussian-mixture components: no change (blue) and change (orange).

TABLE I
PRETRAINING RUNS. STEPS FOLLOW FROM THE DATASET ITEMS PER EPOCH.

	MoCo	SeCo	TeCo	CACo
Items per epoch	20,000	2,000	4,000	4,000
Optimizer steps	31,200	2,800	6,000	6,000
Training time (GPU-h)	23.9	4.6	9.2	9.1

III. METHODOLOGY

A. Pretraining data

We pretrain on the public CACo archive `clean_10k_geography`, which contains 2,000 locations near populated places, each imaged on ten Sentinel-2 [13] dates: five in 2017–18 and five in 2021–22 (Fig. 1). We use the RGB bands (B4, B3, B2) scaled with the authors' fixed quantiles. The archive's name appears to count images per period ($2{,}000 \times 5$). The step counters in the authors' released "100k" checkpoints imply 20,000 locations, ten times ours; the paper trains on that dataset for 1,000 epochs.

B. Objectives

All four objectives use the authors' reference implementation [14] unchanged; our code wraps it with logging, checkpointing and export. Each training item yields a query and keys. Negatives come from a per-subspace queue; SeCo's subspaces 1–2 and all TeCo and CACo subspaces also use other keys in the batch (MoCo v2 and SeCo's subspace 0 use only the queue).

- **MoCo v2:** two augmentations of one image, one embedding space. Each of the 20,000 images is an item.
- **SeCo:** three dates of a location drawn from all ten, giving three subspaces as in [2]. Each location is one item.
- **TeCo** (the paper's "long-term temporal contrast"): the three dates are drawn, with replacement, from a single period, and the same location's views from the other period are added as negatives. Each (location, period) pair is an item.
- **CACo:** TeCo with change-awareness; its three dates are drawn without replacement. For every item, the ratio of the long-term to the short-term feature distance under the momentum encoder is smoothed with an exponential moving average (weight 0.8 on history). After every epoch a two-component Gaussian mixture splits the items into change and no-change; for no-change items, the long-term key of the same subspace is masked out of the softmax during the next epoch. Fig. 2 shows the estimated ratios.

C. Training

Every encoder is a ResNet-18 trained on one GPU with SGD (learning rate 0.03, momentum 0.9, weight decay 10^{-4}), batch size 256, 128-d embeddings, temperature 0.07 and encoder momentum 0.999, for 400 epochs with the learning rate divided by 10 at epochs 250 and 350. Each subspace keeps a queue of 4,096 negatives (the paper uses 16,384). Table I lists the resulting optimizer steps and training time, and Fig. 3 the loss curves.

D. Evaluation

We evaluate frozen encoders on EuroSAT [11] (Sentinel-2, 10 classes, the SeCo/CACo split of 16,200 training and 5,400 test images) and on UC Merced [15] (aerial imagery at about 0.3 m per pixel, 21 classes, our stratified split of 1,680/420 images, released with the code). UC Merced is not used in the paper and differs in sensor and resolution, which makes it our distribution-shift stress test.

We compare nine encoders: random initialisation, ImageNet [16] weights, our SeCo, TeCo and CACo, and the authors' released 100k MoCo, SeCo, TeCo and CACo checkpoints. As in the authors' EuroSAT pipeline, images enter as `ToTensor()` tensors (UC Merced resized to 256×256). Unlike their probe, which lets BatchNorm statistics adapt during training, we keep the backbone in evaluation mode so that features are fixed and can be cached. A linear classifier is trained on the 512-d features with Adam (learning rate 10^{-3}) for 100 epochs, decayed by 10 at epochs 60 and 80, batch size 256. We report final-epoch test accuracy averaged over five probe seeds, plus k-NN (k=5, cosine) accuracy; retrieval

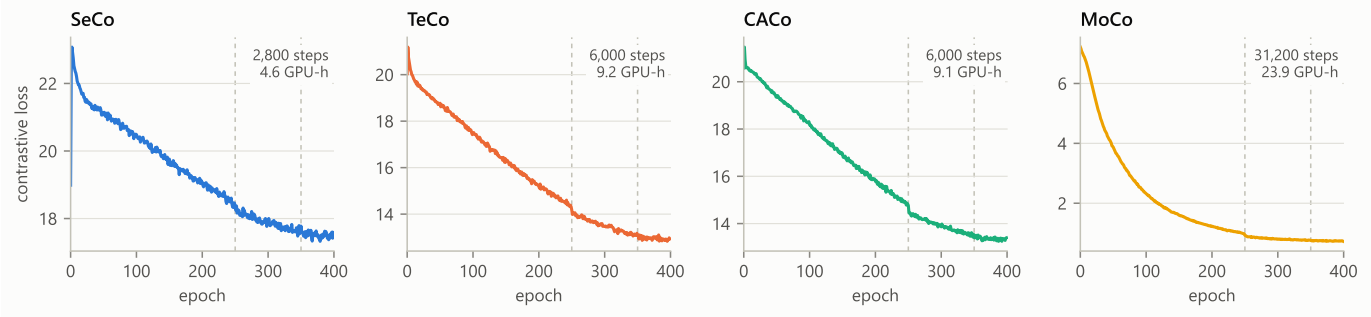


Fig. 3. Pretraining loss per epoch. Dashed lines mark the learning-rate drops. Losses are not comparable across panels because the objectives have different numbers of subspaces and negatives.

precision P@20 is also computed and released with the code. Differences between encoders are tested with a paired bootstrap over test images (10,000 resamples) [17]. The intervals resample test images, using each image's correctness averaged over the five probe seeds; each objective was pretrained once.

E. Hypotheses

Before the final evaluation we wrote four hypotheses with decision rules (released as `docs/HYPOTHESES.md`). H1, H2 and the released-checkpoint part of H4 concern models we had not yet evaluated with this protocol; H3 turns an observation from our first single-seed evaluation into a test.

- **H1** (paper Table 6): on EuroSAT, the authors' checkpoints order SeCo $<$ TeCo $<$ CACo. If the paper's ablation is robust, it should survive a different probe.
- **H2** (paper Table 1): the authors' MoCo v2 checkpoint is below both their SeCo and their CACo, since temporal positives add supervision that plain MoCo v2 lacks.
- **H3** (ablation): at our scale, change-awareness adds less than one point over TeCo on EuroSAT. With few locations, the change estimates rest on noisy distance ratios, so the mask may not separate changed from unchanged places reliably.
- **H4** (stress test): ImageNet's lead over the best satellite encoder of each scale is larger on UC Merced than on EuroSAT, because features tuned to 10 m Sentinel-2 textures should transfer less well to 0.3 m aerial imagery than generic ImageNet features.

IV. Results

Table II lists linear-probe and k-NN accuracy for every encoder. Across five probe seeds the standard deviation is at most 0.17 points on EuroSAT and 1.03 on UC Merced, so differences between encoders reflect the features rather than probe randomness.

A. Claim verification

Table III tests five quantitative claims. Numbers in the paper come from its own probing setup, so we compare orderings and gaps, not absolute values.

The paper's component chain reproduces with the released checkpoints: SeCo $<$ TeCo $<$ CACo on EuroSAT (84.32, 85.77, 86.83), both steps are resolved, and both intervals contain the paper's increments (Fig. 4). CACo also beats the authors' MoCo v2 checkpoint by 3.5 points, but SeCo's lead over it is not resolved, so H2 is unresolved under its decision rule. Two claims do not transfer to our probe. ImageNet features are statistically tied with the best satellite encoder on EuroSAT, and they rank first on BigEarthNet in our earlier run of a 10%-label MLP probe (26,969 training patches; 123,648 validation patches scored), although CACo still beats SeCo there.

TABLE II
Frozen-feature accuracy (%). Linear: final epoch, mean ± s.d. over five probe seeds. k-NN: k=5, cosine.

		EuroSAT		UC Merced	
	Encoder	Linear	k-NN	Linear	k-NN
–	Random	53.00 ± 0.13	42.98	24.14 ± 1.03	49.29
	ImageNet	**87.45** ± 0.04	79.74	**91.62** ± 0.20	**87.86**
Ours 2k loc.	SeCo	67.42 ± 0.05	66.31	58.24 ± 0.80	54.76
	TeCo	72.23 ± 0.06	70.37	69.05 ± 0.41	67.14
	CACo	72.81 ± 0.17	70.69	67.43 ± 0.68	67.38
Authors 20k loc.	MoCo v2	83.37 ± 0.03	80.48	85.76 ± 0.39	87.14
	SeCo	84.32 ± 0.04	83.69	82.95 ± 0.49	84.29
	TeCo	85.77 ± 0.06	83.63	84.19 ± 0.27	84.76
	CACo	86.83 ± 0.08	**85.09**	85.19 ± 0.35	82.62

B. Ablation at small scale

Our own pretraining isolates the same two components on one tenth of the data. Long-term contrast gives a large gain: TeCo beats SeCo by 4.81 points on EuroSAT and 10.81 on UC Merced (Fig. 4). Change-awareness gives no detectable gain: CACo − TeCo is $+0.58$ $[-0.28, 1.43]$ on EuroSAT and -1.62 $[-5.05, 1.81]$ on UC Merced, so H3 is supported. With the authors' checkpoints the same step is resolved ($+1.06$). In the reference code CACo also draws its dates without replacement, so this step measures change-awareness together with that sampling change, at both scales.

Two further observations bear on these comparisons. First, equal epochs are not equal training: TeCo and CACo see every location once per period, so they take 6,000 optimizer steps against SeCo's 2,800 (Table I). The released checkpoints

TABLE III

CLAIMS FROM THE PAPER (RESNET-18) AND OUR TESTS. OUR GAPS ARE PAIRED DIFFERENCES IN LINEAR-PROBE ACCURACY WITH 95% BOOTSTRAP INTERVALS (BIGEARTHNET: ONE RUN OF AN MLP PROBE, MICRO-AVERAGED AVERAGE PRECISION, COMPARED AS POINT ESTIMATES). "HOLDS": THE PAPER'S DIRECTION IS REPRODUCED AND THE INTERVAL EXCLUDES ZERO FOR EVERY COMPARED MODEL; "PARTLY": THIS HOLDS FOR SOME OF THEM; "NOT REPRODUCED": FOR NONE.

Claim	Paper	Ours (authors' checkpoints)	Verdict
CACo improves on SeCo on EuroSAT	91.42 → 93.08, +1.66 (Tab. 6)[a]	84.32 → 86.83: +2.51 [1.64, 3.39]	Holds
Long-term contrast, then change-awareness, each add accuracy (H1)	+1.14, then +0.52 (Tab. 6)	+1.45 [0.59, 2.32], then +1.06 [0.26, 1.89]	Holds
Satellite objectives beat MoCo v2 on satellite data (H2)	87.22 < 90.05, 93.08 (Tab. 1)[b]	CACo − MoCo +3.46 [2.49, 4.44]; SeCo − MoCo +0.94 [−0.05, 1.95]	Partly
Satellite pretraining beats ImageNet on EuroSAT	86.16 < 93.08 (Tab. 1)	ImageNet − CACo +0.61 [−0.37, 1.61]	Not reproduced
CACo beats SeCo and ImageNet on BigEarthNet (10% labels)	67.89 > 65.80, 65.43 mAP (Tab. 2)	CACo 63.18 > SeCo 60.57; ImageNet 66.17	Partly

[a]Tab. 6's SeCo with CACo's geographic sampling, which we take to match the released `_geo_` checkpoints; Tab. 1 reports 90.05 → 93.08 (+3.03) for SeCo with its original sampling.

[b]The released MoCo v2 checkpoint was trained for 200 epochs; the paper trains its 100k baselines for 1,000.

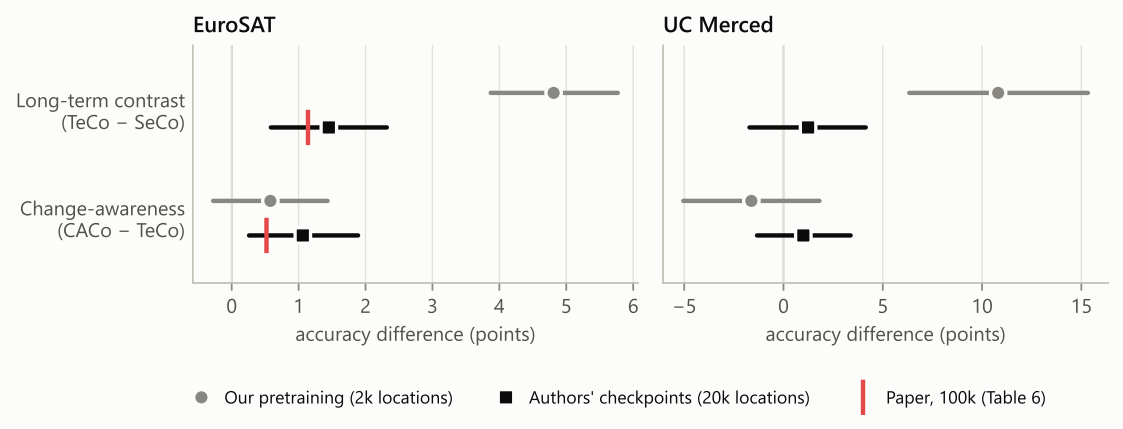


Fig. 4. Contribution of each CACo component, as paired accuracy differences with 95% bootstrap intervals. Red marks are the paper's Table 6 increments (EuroSAT, 100k data).

show the same imbalance: their BatchNorm step counters read 156,000 for TeCo and CACo and 78,000 for SeCo (2:1, against 2.1:1 in our runs). Part of the long-term gain at either scale may therefore come from longer training. Second, our MoCo v2 encoder, trained under the same settings but iterating over every image (31,200 steps), scored 79.48% on EuroSAT and 84.29% on UC Merced in our original single-seed evaluation (on a different UC Merced split from Table II). That beats all three temporal objectives at our scale. With the released checkpoints, where MoCo v2 had a comparable number of steps (156,200), CACo is ahead on EuroSAT (+3.46) and level with it on UC Merced (−0.57 [−3.33, 2.24]). Training length, not just the objective, has to be controlled when comparing these methods.

C. Stress test: UC Merced

UC Merced images have about 30 times finer resolution than Sentinel-2, a regime the paper names as a limitation. As predicted (H4), ImageNet's lead over the best satellite encoder grows on it: from 0.6 to 5.9 points for the released checkpoints, where MoCo v2 is the best satellite encoder on UC Merced (85.76%), and from 14.6 to 22.6 points over our best temporal encoder. Our MoCo v2 encoder, measured only in the original single-seed evaluation, does not follow this pattern: ImageNet's lead over it falls from 8.0 to 5.7 points. The released temporal checkpoints keep their EuroSAT ordering on UC Merced, but neither step is resolved (+1.24 and +1.00); at our scale CACo falls slightly below TeCo (−1.62, within noise). Fig. 5 shows test images that ImageNet features classify correctly and the CACo checkpoint does not. Some of these errors are between related classes (dense vs. medium residential, chaparral vs. forest); others confuse distinct objects (storage tanks, parking lots and intersections predicted as airplanes).

D. What the paper doesn't tell you

The probe protocol moves the numbers. Table IV probes the same released checkpoints in two ways. With ImageNet-normalized inputs, BatchNorm statistics that adapt during probe training and batch size 64, CACo (93.31%) and random initialisation (62.80%) come close to the paper's 93.08% and 64.21%, while SeCo (93.06%) exceeds the paper's SeCo. With a frozen backbone and raw inputs, random initialisation drops by almost ten points and the pretrained satellite encoders by

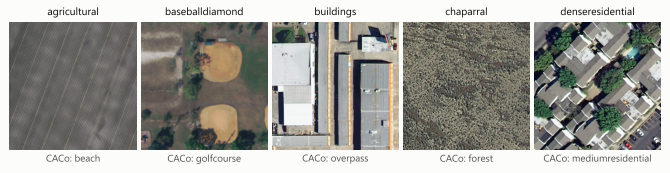


Fig. 5. UC Merced test images classified correctly from ImageNet features but misclassified from the authors' CACo checkpoint in every probe seed (37 of 420). Titles: true class; below: CACo's prediction.

TABLE IV
EuroSAT accuracy of the same encoders under two probe protocols. BN-adapt: ImageNet-normalized inputs, BatchNorm statistics updated during probe training, batch 64, final epoch, one seed. Frozen: Table II.

Encoder	Paper	BN-adapt	Frozen
Random	64.21	62.80	53.00
ImageNet	86.16	85.26	87.45
SeCo 100k	91.42†	93.06	84.32
CACo 100k	93.08	93.31	86.83
CACo − SeCo	+1.66	+0.25	+2.51

†Tab. 6 (SeCo with CACo's sampling); Tab. 1 lists 90.05.

6.5 to 8.7, and ImageNet moves from below both satellite encoders to above them (its lead over CACo is within noise). The CACo − SeCo gap is $+0.25$ under one protocol and $+2.51$ under the other, against $+1.66$ in the paper. The encoders were pretrained on ImageNet-normalized inputs, so feeding raw inputs to a frozen backbone shifts its input distribution; re-estimating BatchNorm statistics is one way to absorb that shift. Comparisons across papers are only meaningful under one protocol.

SeCo gets half the training. The paper notes that TeCo and CACo batches hold half as many locations, but it does not report that this doubles their optimizer steps relative to SeCo, or control for it. Our runs (2.1:1) and the released checkpoints (2:1) both show this imbalance, so the paper's long-term-contrast increment mixes a new objective with longer training.

The change mask is partial. In the reference code, an item judged unchanged loses one long-term key per subspace: the key from that subspace. In subspace 0 this is its only long-term negative, but in the other two subspaces the other subspaces' keys for the other period remain negatives. Together with the sampling difference above, CACo and TeCo differ in two ways, and the mask is narrower than the paper's description suggests. In our run the ratio distribution is unimodal with a right tail (Fig. 2); the mixture model labels the tail as changed, so most items are treated as unchanged.

"10k" and "100k" appear to count images per period. The public 10k archive holds 2,000 locations with five images per period, and the released 100k checkpoints imply 20,000 locations.

V. Challenges

Compute. Each pretraining run used one cluster GPU: 4.6 hours for SeCo, about 9 for TeCo and CACo, and 23.9 for MoCo v2, which iterates over all 20,000 images every epoch. Because the reference code pins PyTorch 1.7 and PyTorch Lightning 1.1.8, we kept a separate legacy environment for training and ran all evaluation in a current one.

Weak training signal. The four losses are not comparable with one another (Fig. 3), so the only way to know whether a run had worked was to probe it downstream. We therefore built a feature-caching probe harness that re-evaluates an encoder in about two minutes on a laptop CPU.

Data access. The original download links for BigEarthNet and UC Merced no longer resolve. We used the official Zenodo release of BigEarthNet [12] and the TorchGeo [18] mirror of UC Merced, and we release our UC Merced split.

Inconsistent early numbers. Our first EuroSAT probes, run on Colab and on the cluster, gave the same randomly initialised network 62.8% and 53.1%. Tracing this to the probe protocol (Table IV) led us to evaluate every encoder with a single harness, five seeds and paired tests.

VI. Ethics

Surveillance. Our pretraining uses Sentinel-2 imagery at 10 m per pixel, where people, vehicles and most individual houses are not resolvable; the UC Merced stress test uses public 0.3 m aerial imagery of land-use scenes. Representations trained to notice change could still be applied to higher-resolution imagery to monitor properties, movements or gatherings, and that use deserves oversight.

Geographic bias. The pretraining locations are sampled around the world's most populated cities. The features are therefore tuned to urban growth and temperate agriculture, and may serve informal settlements, smallholder farms or arid and polar regions poorly. These are often the places where remote sensing could help most.

Compute. Our four pretraining runs logged 46.8 GPU-hours. The main evaluation (Table II) ran on a laptop CPU in under an hour by caching features.

VII. Reflection

Overall. The project delivered what we set out to test: the paper's ablation reproduces with its released checkpoints, while at small scale the change-awareness component shows no measurable effect.

Goals. Our base goal was to pretrain SeCo and CACo with the reference code and show they beat random initialisation; we did, by 14 to 20 points on EuroSAT. Our target goal was a controlled ablation of CACo's components (TeCo) and a check of the paper's claims against its checkpoints; we did both. Our stretch goal, a distribution-shift test on a dataset the paper does not use, became the UC Merced evaluation.

Did it work as expected? Partly. We expected change-awareness to matter most when data is scarce; instead it was undetectable at our scale and resolved only with the authors' checkpoints, which used ten times more locations and longer training. We did not expect plain MoCo v2 or ImageNet features to be so competitive, or the probe protocol to move results as much as the choice of objective.

How the project evolved. We began by trying to match the paper's numbers. When our first evaluations disagreed with each other, we shifted to controlled comparisons: one harness for all encoders, several probe seeds, paired tests, and hypotheses written down before the final evaluation. If we started over, we would fix the evaluation protocol first, log optimizer steps for every objective, and save per-image predictions from the very first run.

With more time we would match optimizer steps across objectives, add pretraining seeds to measure run-to-run variance, evaluate change detection on OSCD, where CACo reports its largest gains, and try a soft change mask using the mixture posterior.

Takeaways. Controls matter as much as models: equal epochs are not equal compute, and evaluation protocol can outweigh architectural choices. Strong simple baselines (MoCo v2, ImageNet) belong in every table. Ablations should ask what each component buys, not just whether the full method wins.

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